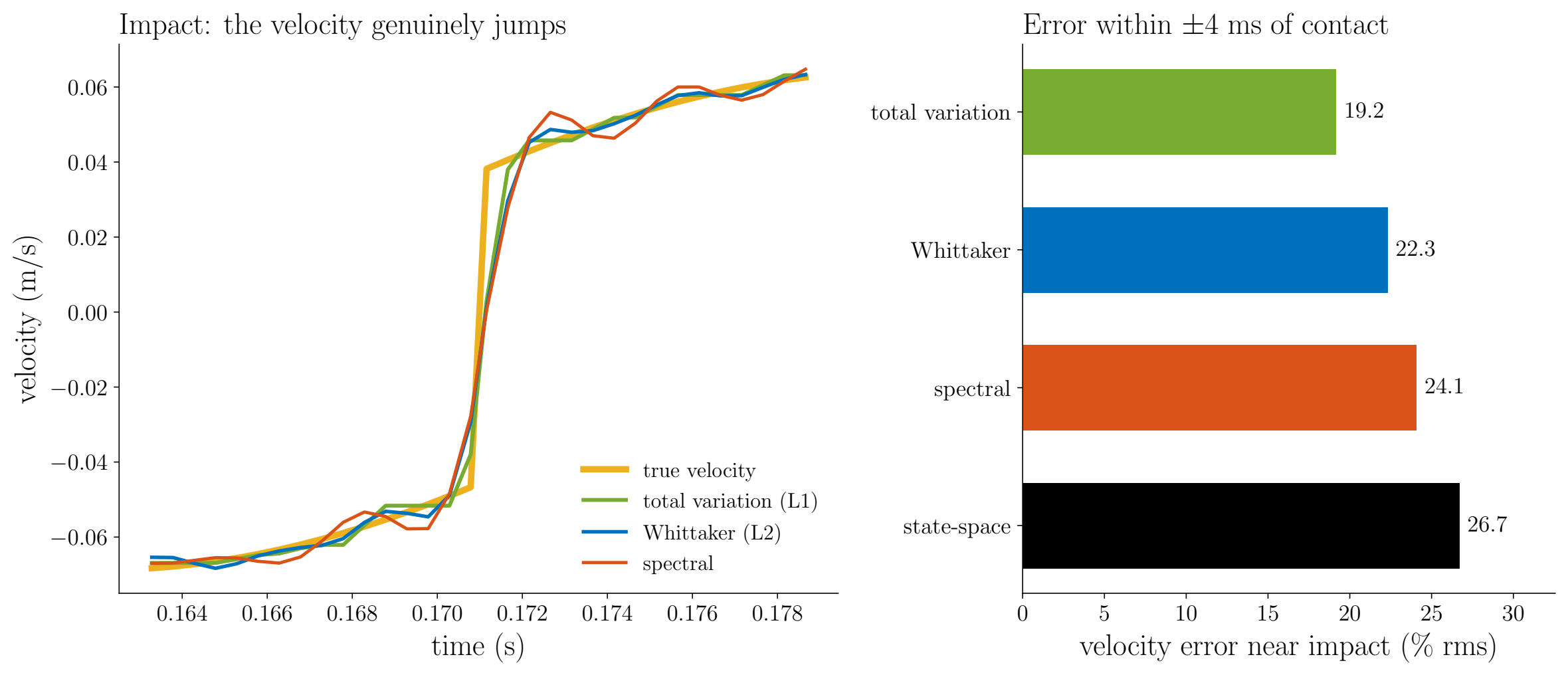
Impact: the velocity genuinely jumps
velocity (m/s)
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
0.164
0.166
0.168
0.170
0.172
0.174
0.176
0.178
time (s)
true velocity
total variation (L1)
Whittaker (L2)
spectral
Error within ±4 ms of contact
total variation
Whittaker
spectral
state-space
19.2
22.3
24.1
26.7
0
5
10
15
20
25
30
velocity error near impact (% rms)